

SECURING THE DIGITAL MINE: A METAHEURISTIC-OPTIMIZED DEEP LEARNING FRAMEWORK FOR INTRUSION DETECTION IN IoT-ENABLED MINERAL OPERATIONS

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Executive Summary: Modern Mining 4.0 integrates hundreds of thousands of Industrial IoT telemetry sensors and SCADA PLCs to automate SAG mills, tailings dams, and ventilation grids. However, IT/OT convergence has eliminated physical air-gaps, exposing unauthenticated industrial protocols (Modbus TCP, DNP3, OPC-UA) to cyber-physical intrusions. Traditional intrusion detection systems (IDS) fail at remote African extraction sites due to excessive feature dimensionality (41+ features), high false alarms, and excessive latency (>150ms) that violates SCADA 20-100ms control loop deadlines. Guided by Design Science Research (DSR), this project engineered an edge-deployable IDS integrating a **Binary Whale Optimization Algorithm (BWOA)** with a **spatial-temporal CNN-LSTM** classifier and **Float16 TFLite quantization**. The artifact achieves **75.61% feature reduction** (10 features), **70.56% multi-class accuracy** on KDDTest+, **96.89% benign precision**, **89.04% DoS recall**, and executes in **0.76 ms** on a 1GB RAM Raspberry Pi 4B (**207x faster** than baseline), satisfying all SCADA control loop deadlines.

1. Problem Context, Industrial Cybersecurity Gap & Solution Objectives (DO1-DO7)

Cyber-Physical Threats in Mining: Unlike IT where data privacy is paramount, mining OT enforces human safety and availability. Compromising Modbus coils on dewatering pumps floods shafts in minutes; spoofing toxic gas sensors (CO, CH4, H2S) suffocates miners; tampering with SAG mill cooling causes catastrophic motor burnout. Unplanned downtime costs ,000-,000/hr (Dragos, 2024).

Edge Hardware & Network Realities: Remote African concessions (Tarkwa, Obuasi, Katanga) operate over solar arrays and intermittent satellite links. Gateway nodes have strictly constrained compute (1GB RAM Raspberry Pi 4B). Cloud-reliant security architectures are physically and economically unviable.

- The 7 Quantitative Design Objectives (DO1-DO7):**
- DO1 (Latency):** Single-sample inference < 1.0 ms on 1GB RAM Pi 4B (< 100ms SCADA ceiling).
 - DO2 (Model Size):** Storage footprint < 1.0 MB via Float16 quantization.
 - DO3 (Accuracy):** Multi-class accuracy >= 65.0% across 5 attack classes on KDDTest+.
 - DO4 (Pruning):** Dimensionality reduction >= 70.0% using BWOA wrapper optimization.
 - DO5 (Accuracy Floor):** Enforce >= 75.0% RF CV accuracy during feature selection.
 - DO6 (Edge Autonomy):** Zero internet/cloud dependencies during edge inference.
 - DO7 (Deployment):** Deployable in < 5 minutes via automated CLI sniffer package.

2. Design Science Research (DSR) Lifecycle & Four-Layer Microservice Architecture

This project strictly implements Peffers et al. (2007) and Hevner et al. (2004) DSR guidelines across six iterative lifecycle phases: Problem Identification -> Objectives Definition -> Design & Architecture -> Development & Demonstration -> Empirical Evaluation -> Dissemination.

Layer	Subsystem Name	Technology Stack	Operational Function & Responsibilities	Key Metric
Layer 1	Edge Ingestion Sniffer	Node.js 20 / Libpcap	Promiscuous packet capture; parses Modbus:502, DNP3:20000, OPC-UA:4840; streams 10 BWOA JSON metrics at line speed.	0.33-0.48ms
Layer 2	BWOA Optimization	Python 3.11 / NumPy	Binary Whale Optimization with V-shaped transfer function; prunes 41 features to 10 with 75% accuracy floor penalty.	75.61% Pruned
Layer 3	CNN-LSTM Classifier	TensorFlow 2.15 / TFLite	Conv1D spatial filter + LSTM temporal tracking + Float16 post-training quantization for ARM Cortex-A72 execution.	0.76ms / 0.82MB
Layer 4	Operational SaaS UI	Laravel 12 / FastAPI	FastAPI async microservice (port 8001) + Multi-tenant Livewire dashboard with live alert feeds and incident logs.	<1s UI Stream

Relevant Visual Assets for Presentation Slide 1 / Section 1:

- * research/figures/system_architecture.png (End-to-End 4-Layer Edge-to-Cloud Pipeline)
- * research/figures/mining_scada_flowchart.png (Cyber-Physical Mining SCADA Circuit & Defense Boundary)
- * research/figures/dsr_framework.png (6-Stage Design Science Research Lifecycle Diagram)

2. Mathematical Formulations, Deep Learning Architecture & Benchmark Results

Detailed mathematical mechanics of BWOA feature selection, CNN-LSTM neural layers, and classification evaluation on KDDTest+ (22,544 samples).

1. Binary Whale Optimization Algorithm (BWOA) Mathematical Engine

BWOA models feature selection in discrete binary space {0, 1}⁴¹. Whale search agents update positions via four equations:

1. Shrinking Encircling: $D = |C * X^*(t) - X(t)|$, $X_cont(t+1) = X^*(t) - A * D$ (where $A = 2a*r1 - a$, $C = 2*r2$, a decays 2 to 0)
2. Spiral Bubble-Net: $X_cont(t+1) = |X^*(t) - X(t)| * \exp(b*l) * \cos(2*\pi*i) + X^*(t)$ ($b=1.0$ spiral constant, l in $[-1, 1]$)
3. V-Shaped Binary Transfer Function: $V(x_d) = |x_d / \sqrt{1 + x_d^2}|$; $X_d(t+1) = 1 - X_d(t)$ if $\text{rand}() < V(x_d)$ else $X_d(t)$
4. Constrained Multi-Objective Fitness: $\text{Fitness}(X) = \alpha * (1 - \text{Acc}(X)) + (1-\alpha) * (|X|/D) + \text{Penalty}(X)$, where $\alpha=0.3$, $\text{Penalty}=1.0$ if $\text{Acc} < 0.75$ or $|X| < 10$.

Rank	Feature Name	Gini Score	Category	OT / Mining SCADA Semantic Meaning
1	src_bytes	0.2451	Volume	Detects high-volume DoS packet floods targeting PLC Modbus queues.
2	service	0.1982	Protocol	Maps industrial ports: Modbus:502, DNP3:20000, OPC-UA:4840 vs HTTP:80.
3	flag	0.1420	TCP State	Identifies abnormal SYN floods (S0), teardowns (RSTR), or rejections (REJ).
4	serror_rate	0.1185	Error Rate	Percentage of SYN errors; primary signature of synchronized DoS attacks.
5	same_srv_rate	0.0894	Pattern	Concentration on single service; flags repeated unauthorized command injection.
6	diff_srv_rate	0.0652	Pattern	Dispersion across services; uncovers port scanning and lateral reconnaissance.
7	dst_host_diff_srv_rate	0.0521	Host Pattern	Target host service divergence; isolates subnet sweeps across sensor gateways.
8	protocol_type	0.0412	Layer 3/4	Partitions TCP (SCADA control), UDP (sensor broadcasts), and ICMP streams.
9	hot	0.0278	Access	Sensitive directory access count; isolates Remote-to-Local (R2L) exploits.
10	su_attempted	0.0205	Privilege	Root escalation flag; isolates User-to-Root (U2R) administrative takeovers.

2. Comprehensive Model Benchmarking & Multi-Class Evaluation on KDDTest+

Architecture & Model	Dataset Partition	Features	Accuracy	Macro F1	AUC-ROC	Latency (Pi 4B)	Size (MB)	DO Status
Full CNN-LSTM Baseline (Float32)	NSL-KDD Test	41 Feat	77.70%	0.7571	0.9359	157.66 ms	1.86 MB	FAIL (>100ms)
BWOA CNN-LSTM (Keras Float32)	NSL-KDD Test	10 Feat	70.56%	0.7127	0.8471	35.60 ms	4.88 MB	PASS (<100ms)
BWOA CNN-LSTM (Float16 TFLite)	NSL-KDD Test	10 Feat	70.56%	0.7127	0.8471	0.76 ms	0.82 MB	PASS (207x Fast)
SWaT ICS Transfer Learning (TFLite)	SWaT SCADA	51->10	59.95%	0.5966	0.8650	0.12 ms	1.76 MB	PASS (Transfer)

3. Per-Class Precision, Recall & Operational Mining Significance

Attack Class	Precision	Recall	F1-Score	Test Samples	Industrial Operational Interpretation in Mining SCADA
Normal (Benign)	0.9689	0.6839	0.8018	9,711	96.9% precision eliminates false alarms that cause costly /hr production shutdowns.
DoS (Denial of Service)	0.7514	0.8904	0.8150	7,458	89.0% recall intercepts volumetric buffer flooding targeting SAG mill and dewatering PLCs.
Probe (Reconnaissance)	0.5488	0.7080	0.6183	2,421	Detects malicious subnet discovery, Modbus address enumeration, and port sweeps.
R2L (Remote to Local)	0.5971	0.1449	0.2332	2,887	Captures unauthorized remote attempts to overwrite PLC register configurations.
U2R (User to Root)	0.0134	0.3881	0.0258	67	38.8% recall on extreme 259:1 imbalance via balanced class weighting loss penalty.

Relevant Visual Assets for Presentation Slide 2 / Section 2:

- * research/figures/bwoa_convergence.png (BWOA 100-Iteration Fitness History) | feature_importance.png (Gini Importance Ranking)
- * research/figures/cnn_lstm_architecture.png (Conv1D-LSTM Layer Flowchart) | confusion_matrix.png (22,544-Sample Test Matrix)
- * research/figures/roc_auc_curves.png (Multi-Class ROC Curves) | training_curves.png (Loss/Accuracy Convergence)

3. Edge Validation, Economic Impact & Presentation Blueprint

Hardware execution benchmarks, software verification, expert UAT, industrial downtime ROI, and slide-by-slide presentation blueprint.

1. Raspberry Pi 4B Hardware Latency & Resource Utilization Benchmarks

Deployment Target Platform	Quantization Format	Mean Latency	P95 Latency	Throughput	Peak RAM	Power Draw	SCADA Status (<100ms)
Raspberry Pi 4B (1GB RAM)	TFLite Float16	0.76 ms	1.10 ms	1,315 pkt/s	290.31 MB	2.5 W	PASS (131x Margin)
Raspberry Pi 5 (4GB RAM)	TFLite Float16	0.42 ms	0.68 ms	2,380 pkt/s	295.10 MB	3.8 W	PASS (238x Margin)
AWS EC2 t3.medium (Ubuntu)	TFLite Float16	0.18 ms	0.31 ms	5,555 pkt/s	180.20 MB	Cloud Mngd	PASS (555x Margin)

2. Quality Testing & Expert UAT Evaluation:

Automated Test Suite: 75/75 tests passing (100% pass rate in 125.6s) across 9 test suites verifying BWOA math, CNN-LSTM shapes, API routes, metrics, and edge profiling.

User Acceptance Testing (n=5 Experts):

- * Threat Alert Clarity: **4.8 / 5.0** (clear plain-text labels)
- * Dashboard Responsiveness: **4.9 / 5.0** (<1s live stream)
- * CLI Sniffer Setup: **4.7 / 5.0** (2m 14s mean time)
- * Risk Triage Trust: **4.6 / 5.0** | Concession Fit: **4.85 / 5.0**
- * **Composite UAT Score: 4.4 / 5.0 (Strong Accept)**

UN SDG Alignment: SDG 9 (Infrastructure), SDG 8 (Miner Life Safety), SDG 17 (Bilateral Russian-African Academic Partnership).

3. Economic Downtime ROI & Risk Mitigation:

Unplanned SCADA outages cost -/hr. The ,500/yr edge deployment yields massive financial returns:

Autonomous Haulage (24h Outage): ,000 risk vs ,500 IDS -> **200x ROI**

Crushing/Milling SCADA (18h Outage): ,000 risk vs ,500 IDS -> **300x ROI**

Tailings & Ventilation Grid (8h Outage): ,000 financial risk + **Catastrophic Life Safety Risk -> 260x ROI + Life Safety**

Large Concession (50,000 t/day): .16M risk -> **1,389x ROI**

4. Presentation Design Blueprint (Slide-by-Slide & Poster Layout Guide)

Slide / Section	Target Content, Key Messages & Core Takeaways	Recommended Diagrams & Visual Assets
Slide 1: Title & Problem Context	Title, UNESCO Track 3, authors, air-gap loss, Modbus/DNP3 vulnerabilities, SCADA 20-100ms deadlines, -/hr downtime risks.	mining_scada_flowchart.png
Slide 2: DSR Methodology & Architecture	6-Stage DSR framework, 4-layer decoupled architecture (Sniffer -> BWOA -> CNN-LSTM -> Dashboard), database ER schema.	system_architecture.png dsr_framework.png, er_diagram.png
Slide 3: BWOA Feature Selection	Whale hunting math, V-shaped transfer function, 75% accuracy floor penalty, 75.61% feature reduction (41->10 feat), Gini scores.	bwoa_convergence.png feature_importance.png
Slide 4: Neural Model & Quantization	Conv1D spatial + LSTM temporal architecture, Float16 quantization reducing model to 0.82MB, training convergence curves.	cnn_lstm_architecture.png training_curves.png
Slide 5: Benchmark Evaluation	70.56% accuracy on KDDTest+ (22,544 samples), 96.89% benign precision, 89.04% DoS recall, confusion matrix, ROC curves.	confusion_matrix.png roc_auc_curves.png
Slide 6: Edge Hardware & Latency	0.76ms latency on Raspberry Pi 4B (207x speedup vs 157.66ms baseline), 290MB RAM, 2.5W power, SCADA <100ms compliance.	latency_comparison_barchart.png dashboard_wireframe.png
Slide 7: Quality, ROI & Conclusion	75/75 unit tests passed, 4.4/5.0 UAT specialist score, 200x-1389x economic ROI, UN SDGs 8/9/17 alignment, 8 future research directions.	uml_sequence_diagram.png uml_use_case.png